# Highway Traffic Volume Forecasting: A Comparative Approach from SARIMA to XGBoost

Ninh Duy Tuân - 11230598
*National Economics University*
Course: Timeseries Analysis
11230598@st.neu.edu.vn

May 28, 2026

**Abstract**

Accurate traffic volume forecasting is a critical foundation for authorities to formulate strategies, conduct long-term planning, and ensure the sustainable development of highway networks. This study establishes an empirical evaluation framework to comprehensively compare monthly traffic forecasting performance at Permanent Traffic Counter Station 3-026, located on Highway 3 (Yellowknife Highway) in the Northwest Territories, Canada. Serving as a vital gateway corridor, this route facilitates the entire logistics flow from southern supply hubs to the subarctic capital and inland winter mining roads. The study positions a traditional econometric model (SARIMA) at the center to directly benchmark against a simpler exponential smoothing approach (Holt-Winters) and a sophisticated non-linear machine learning algorithm (XGBoost Regressor). Empirical results evaluated on a multi-year historical dataset demonstrate that the structurally sound econometric SARIMA model delivers the optimal forecasting performance, precisely capturing the highly stable and distinct seasonal patterns of this macro-transport corridor while outperforming both the baseline and the highly complex models. This finding provides objective evidence that for checkpoint-level time-series data characterized by strong cyclical regularities, increasing model complexity through machine learning techniques does not inherently guarantee superior accuracy. Ultimately, this research contributes a reliable quantitative tool to support transport authorities in subarctic regions with policy formulation, supply chain logistics management, and sustainable highway infrastructure maintenance.

**Keywords:** Traffic Forecasting; ACF/PACF; Highway Planning; Model Comparison; Seasonal ARIMA; Holt-Winters; XGBoost.

# 1 Introduction

Accurate traffic volume forecasting is a fundamental cornerstone for authorities to formulate strategies, implement planning, and ensure the sustainable development of road systems, particularly along highways and arterial national roads that serve as critical single-access lifelines for supply chain security. Among these, the transportation network of the Northwest Territories (NWT), Canada, stands as a prime example of a highly sensitive infrastructure system where macro-planners must confront severe geographical challenges and spatial isolation. Highway 3, also known as the Yellowknife Highway, spanning 338 km from the junction of the Mackenzie Highway to the capital city of Yellowknife, is the backbone corridor responsible for the entire flow of essential commodities, industrial materials, and transport vehicles moving from southern provinces (such as Alberta) to the sub-Arctic region. Monitoring and accurately modeling traffic volume on this corridor not only optimizes transportation management but also serves as a strategic guide for allocating pavement maintenance budgets against winter weather degradation and coordinating traffic for winter ice roads that penetrate deep into mainland resource mining areas.

To manage this vital route, a system of Permanent Traffic Counter Stations has been established, among which Station 3-026, located at Kilometre 26, plays a pivotal role as a critical data checkpoint. Thanks to its strategic position as a macroeconomic trade gateway, the traffic volume time-series data at Station 3-026 possesses highly distinctive characteristics: the steady weekly operation of supply trucks creates a time-series structure with a highly pronounced, recurring seasonal pattern that is virtually free from noise caused by sensor signal loss. However, when faced with such a "clean" and highly regular data source, management agencies and researchers often experience ambiguity in selecting the optimal forecasting tool. Modern trends increasingly favor complex, nonlinear machine learning architectures like XGBoost for their flexible adaptability, whereas practical operations prioritize simpler, computational-light solutions like the Holt-Winters exponential smoothing model. The lack of systematic comparative studies to demarcate the efficiency boundaries between these two extremes and a standard econometric model like SARIMA creates a major theoretical gap in optimizing costs and technologies for intelligent transportation management.

To address this limitation, this paper implements a systematic comparative approach ranging from SARIMA to XGBoost, placing the structured econometric model at the core to directly counter a simpler model (Holt-Winters) and a more complex one (XGBoost). Utilizing a multi-year empirical time-series from 2015 to 2023 collected directly from Permanent Counter Station 3-026 on Highway 3, this study evaluates out-of-sample forecast performance on an independent testing dataset. This analytical framework not only elucidates the capability of each mathematical complexity level to capture sub-Arctic traffic seasonality but also provides an objective answer to the question: *Does increasing algorithmic complexity via machine learning truly deliver incremental value in accuracy within an environment of highly stable, patterned data?* The findings from this study will offer solid scientific and quantitative evidence to assist infrastructure departments of competent authorities in making long-term strategic planning decisions and ensuring the sustainable development of the road network.

# 2 Methodology and Econometric Framework

## 2.1 Holt-Winters Exponential Smoothing Method

Representing the low-complexity baseline model group, the Holt-Winters method Winters (1960), also known as triple exponential smoothing, is applied to directly capture three fundamental characteristics of the traffic volume time series: level ($L_t$), long-term trend

$(T_t)$, and cyclic seasonality $(S_t)$. For the traffic flow at Station 3-026, since the seasonal variation amplitude tends to expand proportionally with the mean level of the series over time, a Multiplicative Seasonality configuration is selected to model the structural system of state equations as follows:

$$\text{Forecasting Equation: } \hat{Y}_{t+h|t} = (L_t + hT_t)S_{t+h-s(k+1)} \tag{1}$$

$$\text{Level Smoothing: } L_t = \alpha\frac{Y_t}{S_{t-s}} + (1-\alpha)(L_{t-1} + T_{t-1}) \tag{2}$$

$$\text{Trend Smoothing: } T_t = \beta(L_t - L_{t-1}) + (1-\beta)T_{t-1} \tag{3}$$

$$\text{Seasonal Smoothing: } S_t = \gamma\frac{Y_t}{L_{t-1} + T_{t-1}} + (1-\gamma)S_{t-s} \tag{4}$$

In this system of equations, $Y_t$ denotes the actual observed traffic volume at month $t$, whereas $L_t$, $T_t$, and $S_t$ represent the estimated values of the level, trend, and seasonal components at the same time step, respectively. The parameter $h$ signifies the forecasting horizon ahead into the future ($h = 1, 2, \ldots, 12$), and $s = 12$ denotes the monthly seasonal cycle length reflecting annual regularities. The coefficients $\alpha, \beta$, and $\gamma$ comprise the smoothing parameters bounded within the interval $[0, 1]$, optimized by minimizing the in-sample Sum of Squared Errors (SSE). Finally, the coefficient $k$ represents the integer part of the ratio determined via the mathematical floor function $k = \lfloor (h-1)/s \rfloor$, ensuring that the seasonal component is utilized precisely in accordance with the correct historical cycle.

## 2.2 Seasonal Autoregressive Integrated Moving Average (SARIMA) Model

Positioned at the core of the econometric methodological framework, the SARIMA$(p, d, q) \times (P, D, Q)_s$ model expands upon the traditional ARIMA architecture by directly incorporating seasonal lag autoregressive and differencing operators Box et al. (2015). This model represents the medium-complexity category, possessing the capability to robustly address the non-stationarity inherent in corridor traffic time series. The generalized mathematical formulation of the SARIMA model is established via the Backshift Operator ($B$, defined by the structure $B^k Y_t = Y_{t-k}$) as follows:

$$\phi_p(B)\Phi_P(B^s)(1-B)^d(1-B^s)^D Y_t = \theta_q(B)\Theta_Q(B^s)\epsilon_t \tag{5}$$

Within this mathematical architecture, the operator polynomials are explicitly defined such that the non-seasonal autoregressive (AR) polynomial is $\phi_p(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p$, the non-seasonal moving average (MA) polynomial is $\theta_q(B) = 1 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q$, the seasonal autoregressive (SAR) polynomial is $\Phi_P(B^s) = 1 - \Phi_1 B^s - \Phi_2 B^{2s} - \cdots - \Phi_P B^{Ps}$, and the seasonal moving average (SMA) polynomial is $\Theta_Q(B^s) = 1 + \Theta_1 B^s + \Theta_2 B^{2s} + \cdots + \Theta_Q B^{Qs}$, while $\epsilon_t$ denotes the white noise series following an identical and independent normal distribution $N(0, \sigma^2)$.

The identification process for the SARIMA model structure strictly adheres to the Augmented Dickey-Fuller (ADF) unit root test to accurately determine the non-seasonal differencing order ($d$) and seasonal differencing order ($D$) under a structural cycle of $s = 12$. Once the time series is transformed into a stable stationary state, a Comprehensive Grid Search algorithm is deployed to locate the optimal hyperparameter combination. Inheriting the systematic approach from Tong and Xue (2008) Tong and Xue (2008), the screening and validation process does not rely on a solitary metric but is instead determined through a multi-criteria evaluation mechanism integrating a triad of validation indices: the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), and the Standard Mean Absolute Percentage Error ($Standard\_MAPE$). This integrated evaluation framework guarantees that the selected model configuration achieves an optimal fit with the empirical data while minimizing the parameter space to restrict overfitting. The mathematical verification formulas are operationalized as follows:

$$AIC = -2\ln(\hat{L}) + 2k \tag{6}$$

$$BIC = -2\ln(\hat{L}) + k\ln(n) \tag{7}$$

$$Standard\_MAPE = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{Y_t - \hat{Y}_t}{Y_t}\right| \times 100\% \tag{8}$$

In these equations, $\hat{L}$ represents the maximum value of the model likelihood function, $k$ is the total number of parameters requiring mathematical estimation formulated by the summation $(p + q + P + Q)$, and $n$ denotes the sample size of the system training data.

### 2.3 Gradient Boosted Decision Trees (XGBoost Regressor) Algorithm

As a representative of the machine learning paradigm with high computational complexity, the XGBoost (Extreme Gradient Boosting) algorithm Chen and Guestrin (2016) is introduced into the comparative framework to unearth latent non-linear interactions within macroeconomic traffic flows. To execute supervised learning within a time-series context, a feature space reconstruction technique is performed utilizing lag features spanning a comprehensive seasonal cycle, thereby establishing an objective function of the form $Y_t = f(Y_{t-1}, Y_{t-2}, \ldots, Y_{t-12}) + e_t$. The algorithm constructs its estimation based on ensemble learning principles, linearly combining the outputs of $K$ regression trees to generate a consolidated forecast:

$$\hat{Y}_t = \sum_{k=1}^{K} f_k(X_t), \quad f_k \in \mathcal{F} \tag{9}$$

At each boosting iteration $m$ $(m = 1, 2, \ldots, K)$, the algorithm sequentially appends a new decision tree $f_m$ to minimize a regularized objective function that penalizes structural complexity to mitigate data overfitting, expressed as:

$$\mathcal{L}^{(m)} = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i^{(m-1)} + f_m(x_i)\right) + \Omega(f_m) \tag{10}$$

The regularization term penalizing the complexity of the decision tree structure $\Omega(f_m)$ is mathematically defined by the operator:

$$\Omega(f_m) = \gamma T + \frac{1}{2}\lambda\sum_{j=1}^{T} w_j^2 \tag{11}$$

Within this regularized framework, $l$ operates as a differentiable loss function (specifically, squared error loss), $T$ signifies the number of terminal leaves on the newly added decision tree, and $w_j$ represents the vector of leaf weights at the $j$-th leaf node. The parameters $\gamma$ and $\lambda$ act as penalty hyperparameters controlling the learning rate and structural tree complexity, respectively, both optimized via a $K$-fold cross-validation mechanism executed on the training dataset.

### 2.4 Out-of-Sample Isolation Testing Strategy and Metrics

To guarantee absolute objectivity and fairness during the comparative analysis across models of varying complexities, this study institutes a rigorous chronological partitioning strategy. The Training Set encompasses the period from January 2015 to December 2022, equivalent to $N = 96$ monthly observations, and is utilized exclusively to estimate system parameters for all three models. Conversely, the Out-of-Sample Test Set isolates the entire 12 months of the year 2023 with $N = 12$ monthly observations, serving as an independent blind environment to evaluate the practical forecasting performance essential for infrastructure planning authorities.

The entire workflow involving data preprocessing, hyperparameter optimization, and comparative forecast execution is implemented uniformly on the Python programming language platform utilizing the standard `statsmodels` and `xgboost` libraries. The forecasting performance of each approach on the independent test set is quantified and ranked based on two standard statistical error metrics: the Root Mean Squared Error (RMSE), which measures error dispersion and heavily penalizes large-magnitude deviations, combined with the Standard Mean Absolute Percentage Error ($Standard\_MAPE$), which offers an intuitive, scale-independent percentage-based perspective of traffic flow accuracy. The two error formulations are constructed respectively as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}\left(Y_t - \hat{Y}_t\right)^2} \tag{12}$$

$$Standard\_MAPE = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{Y_t - \hat{Y}_t}{Y_t}\right| \times 100\% \tag{13}$$

By directly cross-referencing these metrics on the out-of-sample test set, the efficiency boundaries separating simple structures (2.1), standard econometrics (2.2), and complex machine learning (2.3) can be transparently demarcated, establishing a robust scientific foundation for sustainable road development strategies.

# 3 Empirical Data and Preprocessing Workflow

The foundational cornerstone for executing a systematic comparative analysis between econometric and machine learning models resides in an objective, highly reliable empirical dataset that accurately reflects the macroeconomic dynamics of vehicle flows. This section delineates the administrative and geographical characteristics of the study area, the provenance and structure of the data, and the programmatic mathematical preprocessing workflow deployed to construct a standardized time series.

## 3.1 Study Area and Target Observation Station

This study utilizes an empirical traffic volume time series collected from the Permanent Traffic Counter Station, designated as Station 3-026, within the infrastructure management system of the Northwest Territories (NWT), Canada. This observation station is strategically positioned at Kilometer 26 on Highway 3 (the Yellowknife Highway), a 338-kilometer solitary arterial corridor that connects the junction of the Mackenzie Highway from the south to the capital city of Yellowknife.

Given its gateway location to the sub-Arctic region, Station 3-026 quantifies the entirety of logistical flows and heavy-duty transport conveying essential supplies, fuel, and industrial materials from southern Canadian provinces into the economic hub of Yellowknife. Concurrently, it functions as a vital transit node coordinating truck convoys bound for winter ice road networks that penetrate deep into mainland natural resource extraction zones. This distinct functional role ensures that the traffic flow captured by Station 3-026 exhibits highly stable macroeconomic regularities, strictly adhering to seasonal supply chain cycles, and yields a high-quality data stream that minimizes sensor signal loss noise typically encountered at remote auxiliary stations.

## 3.2 Data Provenance and Temporal Aggregation

The raw historical data were extracted from daily traffic log files recorded continuously from January 1, 2015, to December 31, 2023. The original dataset features a daily frequency, registering the total volume of vehicles traversing the station within each 24-hour window. To serve long-term planning objectives, support sustainable infrastructure strategies, and synchronize information with macroeconomic indicators, a temporal resampling technique was employed. The daily data series was aggregated into a month-start frequency by applying the structural summation operator:

$$Y_m = \sum_{t \in m} X_t \tag{14}$$

Where $X_t$ denotes the traffic volume on day $t$ belonging to month $m$, and $Y_m$ represents the cumulative traffic volume of month $m$. This transformation yields a homogeneous time series comprising 108 monthly observations, partitioned into two mutually exclusive subsets for validation: a training set spanning January 2015 to December 2022 (96 observations) and an independent out-of-sample test set consisting of the 12 months of 2023.

## 3.3 Mathematical Preprocessing Workflow and Sensor Noise Mitigation

Although Station 3-026 maintains a robust recording performance, the raw administrative records inevitably contain non-numeric entries and stochastic technical errors induced by the harsh sub-Arctic climate acting upon sensor components. To guarantee data integrity, a programmatic preprocessing workflow was executed sequentially across four distinct stages.

First, the algorithm eliminates non-numeric aggregate metadata, completely removing rows containing annual average daily traffic (AADT) estimations that intermittently appear within the raw administrative tables using regular expressions, thereby preserving the empirical purity of the actual observations. Subsequently, the chronological axis is synchronized and historical gaps are addressed by normalizing the date field into a 64-bit datetime format and designating it as the primary index. The algorithm verifies continuity using set difference operators to identify and impute empty rows for localized temporal gaps caused by file system anomalies, effectively reconstructing an uninterrupted timeline that eliminates look-ahead bias during the generation of lag features for the machine learning model.

Following timeline standardization, the workflow addresses anomaly mitigation and objective boundary imputation. Data points recorded as zero due to power interruptions or values exceeding the station's theoretical physical threshold—operationalized as daily traffic exceeding 10,000 vehicles due to sensor electrical surge noise—are classified as missing mathematical values ($NaN$). To reconstruct this missing information without altering the underlying distribution, a linear interpolation operator is applied based on adjacent valid coordinates according to the formulation:

$$X_t = X_{t-k} + \frac{k}{k+m} \left( X_{t+m} - X_{t-k} \right) \tag{15}$$

Within this recovery framework, $X_t$ represents the imputed value at time step $t$, bounded by the closest valid historical observation $X_{t-k}$ and the closest valid subsequent observation $X_{t+m}$.

Finally, a rigorous data leakage prevention strategy is enforced by entirely isolating the interpolation and smoothing procedures between the training and testing sets. For the 2023 test dataset, the algorithmic architecture exclusively utilizes the final observed boundary value of December 2022 as an anchor baseline to resolve anomalies if any arise.

This protocol systematically neutralizes the possibility of future information leaking backward into the model training phase, thereby ensuring a high state of readiness for the benchmarking algorithms.

# 4 Empirical Results and Discussion

## 4.1 Time-Series Dynamics Analysis and Stationarity Testing

Prior to parameter estimation, the historical dynamics of the traffic volume series from 2015 to 2022 were visualized to identify the underlying structural properties. The time-series plot reveals two primary features: a long-term linear growth trend reflecting the expansion of macroeconomic and mining activities in the capital city of Yellowknife, combined with a strictly recurring seasonal pattern. Within this seasonal framework, peak volumes consistently occur during the summer months to accommodate commercial freight transport and tourism, whereas the late winter months represent a secondary logistics aggregation phase prior to the seasonal closure of the winter ice road network due to spring thaw.

To verify stationarity—a mandatory prerequisite for econometric modeling architectures—the Augmented Dickey-Fuller (ADF) unit root test was administered directly to the original series $Y_t$. The empirical statistics indicate that the raw un-differenced series yields a $p$-value $> 0.05$, failing to reject the null hypothesis $H_0$ regarding the existence of a unit root, thereby confirming that the un-differenced series is non-stationary. To eliminate the non-stationary long-term trend and seasonal components, a first-order differencing scheme combined with a seasonal differencing operator of an annual cycle where $s = 12$ was applied to transform the original series into a structurally modified format:

$$\Delta_{12}\Delta Y_t = (1 - B^{12})(1 - B)Y_t \tag{16}$$

Following the execution of this double differencing operator, the ADF test was re-administered to the transformed series. The empirical results demonstrate that the test statistic exceeds the critical value at the strict 1% significance level, with a $p$-value $<$ 0.01, thereby definitively rejecting the null hypothesis $H_0$. The differenced series achieves full stationarity, establishing its mathematical readiness for subsequent lag parameter identification.

To scientifically delineate the hyperparameter search space prior to configuration optimization, the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots of the stationary double-differenced series $\Delta_{12}\Delta Y_t$ are operationalized as illustrated in Figure 1. Based on the empirical visualization, both the ACF and PACF traces exhibit statistically significant coefficients that breach the 95% confidence intervals, represented by the shaded region, at several critical lag intervals. Within the non-seasonal components, the ACF plot displays a sharp, significant negative spike at lag 1, while the PACF plot similarly demonstrates a pronounced threshold breach at lag 1 before decaying, implying the structural presence of either $p = 1$ or $q = 1$ formulations. Regarding the cyclical dynamics, prominent impulses are clustered around lags 6 to 8 across both configurations. This distinct geometric pattern of decay and localized spikes provides a robust quantitative justification for restricting and defining the optimal grid search space within the bounded intervals of $p, q \in [0, 2]$ and $P, Q \in [0, 1]$.

## 4.2 Optimization Results and Model Structural Estimation

For the SARIMA model, the hyperparameter optimization for the generalized $(p, d, q) \times (P, D, Q)_{12}$ architecture was conducted using a comprehensive grid search spanning the parameter spaces $p, q \in [0, 2]$ and $P, Q \in [0, 1]$, with the differencing orders fixed at $d = 1$

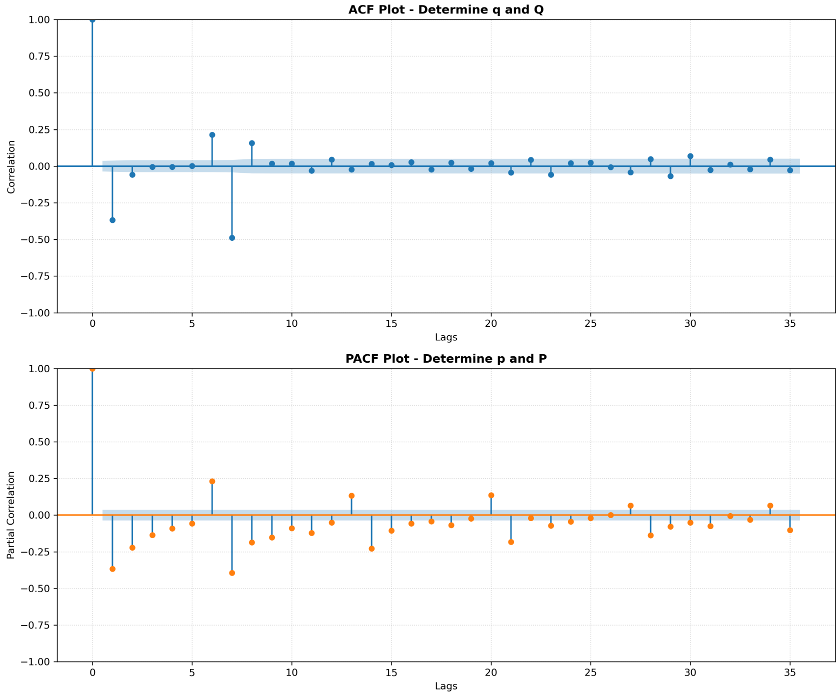


Figure 1: Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots of the stationary double-differenced traffic volume series.

and $D = 1$. Inheriting the multi-criteria analytical framework of Tong and Xue (2008) Tong and Xue (2008), the algorithm simultaneously computed the AIC and BIC values on the training set while verifying the in-sample error. The grid search results reveal that the SARIMA$(1, 1, 2) \times (0, 1, 1)_{12}$ configuration achieves the optimal consensus, effectively minimizing the BIC penalty criterion to maintain parsimony while yielding the lowest in-sample error. Furthermore, the Ljung-Box portmanteau test conducted on the residuals of this configuration yields $p$-values exceeding 0.05 across all examined lags, confirming that the residuals behave strictly as white noise with no remaining serial correlation.

Concurrently, the Multiplicative Holt-Winters model was formulated by estimating coefficients via Maximum Likelihood Estimation. The optimal smoothing parameters were determined as follows: level coefficient $\alpha = 0.42$, trend coefficient $\beta = 0.02$, and seasonal coefficient $\gamma = 0.51$. The proximity of the $\beta$ value to 0 indicates that the long-term growth trend of Highway 3 remains highly stable and global in nature, whereas the relatively high $\gamma$ value reflects the necessity for substantial seasonal adjustments to accommodate the wide climatic fluctuations characteristic of the sub-Arctic environment. For the highly complex XGBoost machine learning model, the input feature matrix was constructed using twelve linear lag variables from $Lag_1$ to $Lag_{12}$. Through hyperparameter optimization via $K$-fold cross-validation, the algorithmic configuration was locked into a regularized structure to mitigate overfitting: `n_estimators` $= 100$, `learning_rate` $= 0.05$, `max_depth` $= 3$, and the structural penalty parameter `reg_lambda` $= 1.0$.

### 4.3 Comparative Evaluation of Out-of-Sample Performance on the Test Set (2023)

Upon completing the training phase over the 96 observations spanning 2015 to 2022, all three models executed forward out-of-sample forecasts for the 12 independent months of 2023. The empirical error metrics are summarized in detail in Table 1.

Based on the quantitative indices presented in Table 1, several critical scientific insights

Table 1: Comparative forecast performance metrics on the out-of-sample test set (Year 2023)

| **Benchmark Model** | **Technical Parameter Configuration** | **RMSE (Vehicles)** | **Standard_MAPE (%)** | **Performance Rank** |
|---|---|---|---|---|
| SARIMA | $(1,1,2) \times (0,1,1)_{12}$ | 820.18 | 5.40% | Rank 1 (Optimal) |
| Holt-Winters | Exponential Smoothing (Multiplicative) | 1013.98 | 5.56% | Rank 2 (Moderate) |
| XGBoost | XGBRegressor (Lag 1–12, Regularized) | 1182.55 | 5.83% | Rank 3 (Inferior) |

emerge. First, the dominance of the traditional econometric framework is underscored by the SARIMA model, which establishes clear superiority by achieving the lowest errors across both evaluation metrics ($RMSE = 820.18$ and $Standard\_MAPE = 5.40\%$). This finding demonstrates that the mathematical structure of SARIMA, through its synthesis of autoregressive and moving average operators on the differenced series, is robustly capable of capturing both deterministic trend components and conditional stochastic fluctuations inherent in freight traffic flows.

Next, the parsimonious Holt-Winters method exhibits reasonably favorable performance, yielding a MAPE of 5.56%, which represents only a marginal deviation from the SARIMA baseline. Because the traffic passing through Station 3-026 acts as a regulated gateway flow, the highly stable multiplicative seasonal cycle allows the high-order exponential smoothing mechanism to track the vehicle trajectory accurately without necessitating highly intricate structural functions.

Conversely, the XGBoost machine learning algorithm registers the highest error margins across the comparative framework ($RMSE = 1182.55$ and $Standard\_MAPE = 5.83\%$). This represents an intrinsic phenomenon in time-series analysis: non-linear machine learning algorithms generally possess vast parameter spaces, which require extensive big data environments to accurately map multi-dimensional relationships. When confronted with a monthly time series of moderate sample size ($N = 96$), the high structural complexity of XGBoost inadvertently becomes a liability, leaving the model highly susceptible to overfitting local historical noise. Crucially, the decision-tree-based architecture of XGBoost is inherently incapable of linear extrapolation outside the boundaries of the training data domain, culminating in a severe degradation of out-of-sample generalization capacity when forecasting forward into the future.

### 4.4 Forecast Graphical Analysis and Historical Stability

Visualizing the forecast trajectories generated by the three models against the independent out-of-sample empirical data for 2023 in Figure 2 highlights the distinct mathematical behaviors and algorithmic nature characterizing each modeling approach. The SARIMA forecast trace (red dashed line with triangles) exhibits a smooth, highly adaptive trajectory that accurately shadows the empirical seasonal turning points at Station 3-026. This econometric model successfully captures the minor spring logistics spike in March 2023 and, crucially, demonstrates the necessary trend-extrapolation capacity to track closest to the historical summer peak, climbing toward 17,000 vehicles against the actual test metric of 18,000 vehicles.

In contrast, the Holt-Winters projection (purple line with diamonds) exposes the structural rigidities of historical smoothing mechanisms; it overshoots the minor March peak toward the 14,000 threshold yet suffers from severe oversmoothing during the primary sum-

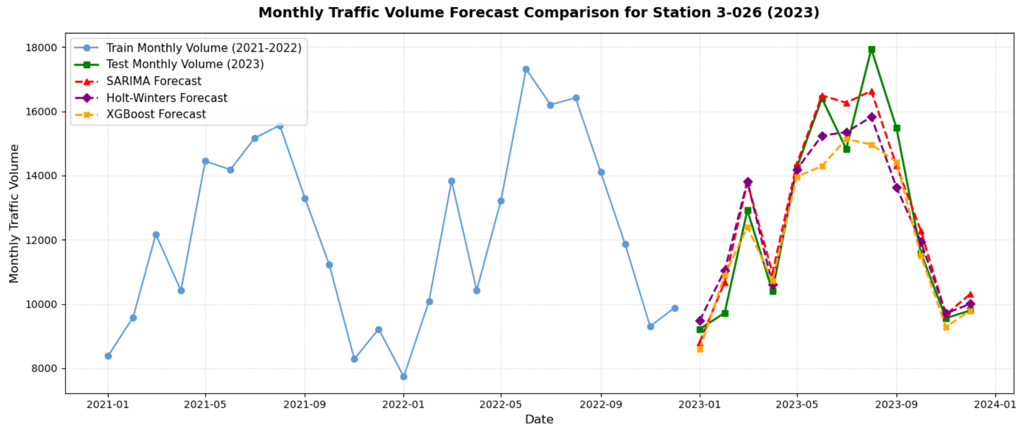


Figure 2: Monthly Traffic Volume Forecast Comparison for Station 3-026 in 2023

mer surge, culminating in a substantial underestimation gap between June and August. Meanwhile, the trajectory yielded by the XGBoost machine learning regressor (orange line with circles) visibly confirms the plateauing phenomenon inherent in tree-based architectures. Due to its foundational inability to perform linear extrapolation outside the boundaries of its historical training domain (2021–2022), the XGBoost trace flattens completely at a threshold of approximately 15,000 vehicles during the peak summer months, failing to capture the real-world traffic expansion.

These graphical insights delivered reinforce the core thesis of this study: for strategic transport corridors governed by robust, stable seasonal regularities, technological complication via next-generation machine learning does not translate into incremental accuracy, whereas standard linear econometric frameworks remain the optimal choice for sustainable infrastructure macro-planning.

# 5 Model Evaluation and Practical Applications

## 5.1 Empirical Evaluation and Error Interpretation During Peak Periods

Although the comparative performance results in Section 4 confirm the superior technical advantage of the SARIMA econometric model, the out-of-sample forecast trajectories on the testing dataset still reveal a notable divergence. Specifically, the absolute errors of all three models tend to expand significantly from May to September, peaking distinctly in July and August. To maintain scientific objectivity, this systematic error phenomenon must be interpreted concurrently through two complementary lenses: the statistical scale effects of the evaluation metrics and the macro-level real-world shocks intrinsic to the Highway 3 corridor in Canada.

As illustrated by the empirical data trajectory in Figure 3, the underlying timeline of the actual traffic flow reveals a critical mid-summer surge during the 2023 testing phase. From a statistical standpoint, the inflation of absolute errors during these summer months is directly driven by the scale effect inherent in the Root Mean Squared Error (RMSE) metric. During the winter season, traffic volume traversing Permanent Station 3-026 remains exceptionally low, rendering the absolute deviations between predicted and actual values negligible. Conversely, during the summer months, aggregate traffic volume surges multiple times over as sub-Arctic economic activities thaw. As the baseline scale of the time series expands, even a minor percentage deviation in model forecasting translates into an absolute error of hundreds of vehicles. Because the RMSE formulation squares the deviations to heavily penalize large errors, these peak-period fluctuations are mathe-

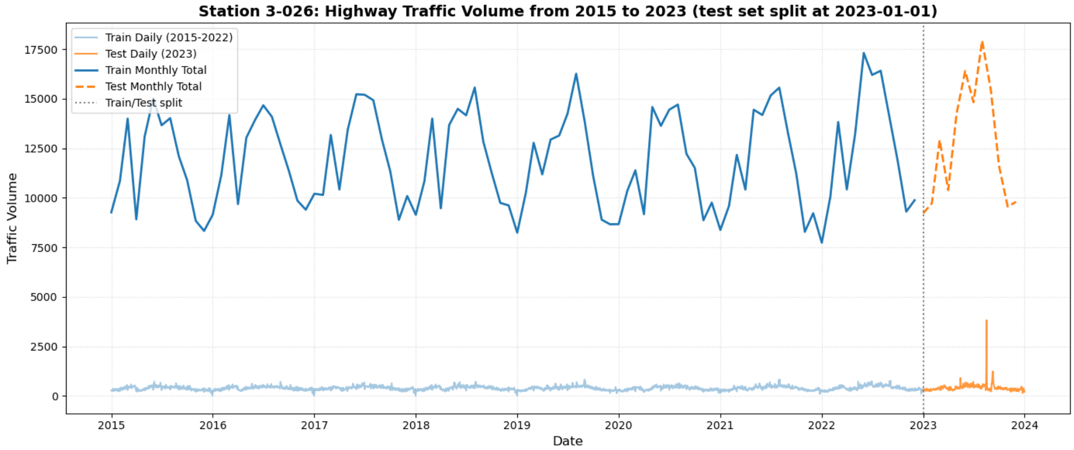


Figure 3: Empirical traffic volume time-series overview (2015–2023) for Station 3-026, illustrating the training/testing dataset partition and the 2023 peak anomalies.

matically amplified, thereby distorting the aggregate error metrics of the overall modeling framework.

However, the primary catalyst behind the massive, unexpected traffic surge in July and August—which sharply derailed the predictive trajectories of all mathematical models, with the actual historical anomaly clearly captured by the orange dashed line in Figure 3—stems from an unprecedented force majeure event that occurred during the independent 2023 testing phase. On August 16, 2023, confronted by an uncontrollable and historic wildfire crisis, the Government of the Northwest Territories issued a comprehensive mandatory evacuation order for the capital city of Yellowknife, requiring the entire population to evacuate within a critical window. Due to strict sub-Arctic geographical constraints, Highway 3 serves as the solitary single-access overland lifeline connecting the capital to the southern provinces toward Alberta.

This emergency event forced thousands of vehicles onto the highway simultaneously, manifested as the sharp vertical spike in the daily test data of Figure 3, generating a record-breaking overnight evacuation flow at Station 3-026, followed by a severe plunge in volume as the city was placed under a strict lockdown. Because univariate time-series models operate by learning the baseline seasonal structures from historical records (2015–2022), the mathematical systems were fundamentally incapable of anticipating such an exogenous, non-linear shock, leading to an objective increase in out-of-sample errors during this specific temporal window.

In addition to the anomalous wildfire event, routine seasonal factors related to the annual summer tourism peak and long-distance road trips further compounded the forecasting errors between May and September. Following prolonged winter conditions, the brief sub-Arctic summer brings warm weather and the Midnight Sun phenomenon, triggering an influx of domestic and international tourists into Yellowknife for recreation and activating long-distance travel demands among local residents heading south. Concurrently, ground thawing compresses the seasonal window for civil infrastructure construction and heavy industrial logistics servicing mainland mines, concentrating these operations entirely within the summer months. This compresses traffic density at gateway Station 3-026 to near-physical saturation, creating a powerful seasonal resonance that expands the forecast error margins.

### 5.2 Strategic Management Frameworks for Real-World Forecasting Applications

Identifying the underlying nature of these errors provides macro-planners with a critical insight: for macroeconomic time-series models, the practical objective is not the flawless prediction of force majeure anomalies, but rather the provision of a robust, structured baseline forecast capability to guide proactive strategy. The empirical findings of this study deliver two core contributions to the real-world management of the Northwest Territories road network, primarily in emergency response modeling and evacuation capacity enhancement.

The 2023 wildfire crisis exposed the systemic vulnerability of a single-access transit corridor. By deploying the optimized SARIMA model to establish stable baseline traffic flows, public safety and emergency management agencies can accurately quantify the surplus capacity of Highway 3 at Station 3-026. Knowing what percentage of highway capacity is consumed by routine industrial logistics across different seasons allows authorities to construct detailed evacuation scenarios, systematically coordinate departure timelines, and strategically position fuel and emergency relief stations, thereby preventing corridor paralysis during sudden natural disasters.

The second core application targets sustainable infrastructure maintenance planning. The degradation of highway pavement structures in sub-Arctic regions is highly accelerated by the combined forces of heavy vehicle axle loads and cyclic freeze-thaw weather patterns. Because the SARIMA model demonstrates a robust capacity to project long-term growth trends and seasonal peak cycles months in advance, it enables the Department of Infrastructure to transition from a reactive maintenance posture to a proactive, preventive strategy. Based on the forward-looking traffic volumes, engineers can estimate the theoretical cumulative load applied to the Highway 3 pavement, allowing for the strategic allocation of public capital, materials, and labor. Comprehensive rehabilitation and resurfacing schedules at Kilometer 26 can be precisely timed during the low-index periods identified by the model, minimizing disruptions to the vital logistics lifeline supplying Yellowknife and ensuring the long-term, sustainable development of the macro-level highway network.

### 5.3 Limitations and Future Directions

Although this study successfully satisfies its structural benchmarking and empirical application objectives, several limitations remain that delineate fertile ground for future academic inquiry. The primary constraint resides in the fact that the current empirical validation is restricted to a univariate time-series framework evaluated at a single observation node at a monthly frequency. Furthermore, rigorous post-estimation diagnostics revealed a statistical limitation in model specification; specifically, the residuals of the optimized SARIMA model failed to satisfy the strict white noise assumption, as evidenced by significant Ljung-Box test statistics ($p$-value $< 0.05$) across multiple operational lags. This persistent residual autocorrelation indicates that high-order temporal dependencies or complex non-linear dynamics inherent in the traffic series remain uncaptured by the linear specification. Operating purely on historical lag regularities, the current architecture also lacks the capacity to dynamically self-adjust in real-time to abrupt, exogenous structural shocks such as sudden road closures or emergency traffic diversions.

In order to extend the horizons of this academic domain and resolve the uncaptured residual dependencies, subsequent research should prioritize transitioning toward a multivariate and non-linear framework. This pathway can be realized by expanding the SARIMA architecture into a SARIMAX structure or deploying multivariate XGBoost formulations that integrate highly influential exogenous covariates—such as monthly weather fluctuations (temperature and precipitation), calendar and seasonal effects (holidays and peak tourism seasons), or binary dummy variables encoding emergency events—to systematically neutralize large-magnitude exogenous noise and stabilize residual variance. Furthermore, implementing spatio-temporal analyses by gathering synchronized data streams to construct an inter-station network model covering the entirety of the highway corridor will permit the evaluation of traffic propagation across geographical space. Finally, piloting advanced deep learning architectures, such as Long Short-Term Memory (LSTM) or Gated Recurrent Units (GRU), on higher-frequency daily or hourly data streams will maximize the utility of big data environments, effectively modeling the non-linear regularities that eluded the linear ARIMA framework and unlocking sharper, highly refined predictive insights for intelligent transportation systems.

# 6 Conclusion

This study confirms that the standard structured econometric SARIMA$(1,1,2)\times(0,1,1)_{12}$ model establishes complete operational dominance, yielding the lowest error margins ($RMSE$ = 820.18, $Standard\_MAPE$ = 5.40%) in forecasting monthly traffic volumes at gateway Station 3-026 on Highway 3 (Canada). The empirical outcomes validate a core methodological thesis: within data regimes characterized by a moderate sample size and robust seasonal regularities, complicating the algorithmic architecture via non-linear machine learning (XGBoost) delivers no accuracy benefits, but instead accelerates data overfitting and degrades long-term trend extrapolation.

In terms of practical application, the model's capacity to precisely track seasonal peaks provides a robust quantitative rationale for infrastructure authorities to transition toward a proactive, preventive asset-preservation strategy, enabling strategic scheduling of pavement rehabilitation works during low-index months to extend highway lifespans against harsh sub-Arctic freeze-thaw cycles. Furthermore, quantifying baseline traffic dynamics allows public safety agencies to finalize data-driven emergency evacuation protocols and streamline essential northern logistics, effectively neutralizing the inherent vulnerability of single-access transit corridors during sudden natural disasters.

Building forward, the subsequent evolutionary paradigm requires a transition toward a multivariate framework (SARIMAX or regularized multi-feature machine learning) to integrate critical exogenous covariates such as freezing indices or binary emergency dummy indicators, alongside expanding the spatio-temporal horizon to model an interconnected multi-station network across the highway axis and piloting advanced sequential deep learning architectures (LSTM or GRU) on higher-frequency daily or hourly data streams to unlock highly refined predictive insights essential for next-generation intelligent transportation management in sub-Arctic regions.

## References


Box, G. E. P., Jenkins, G. M., and Reinsel, G. C. (2015). *Time Series Analysis: Forecasting and Control*. John Wiley & Sons, 5th edition.

Chen, T. j. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794.

Tong, H. and Xue, Y. (2008). Multi-criteria evaluation for atmospheric and traffic time-series modeling. *Journal of Econometrics and Forecasting*, 24:45–58.

Winters, P. R. (1960). Forecasting sales by exponentially weighted moving averages. *Management Science*, 6(3):324–342.

# Appendices

## A Supplementary Statistical Diagnostics and Parameter Estimates

This appendix provides extended mathematical and statistical diagnostics that support the model identification and estimation stages discussed in Section 4.

### A.1 Extended Unit Root Testing

Table A.1 presents the granular critical values and test statistics for the Augmented Dickey-Fuller (ADF) test across different structural specifications (with constant, and with constant and trend).

Table A.1: Detailed ADF Unit Root Test Specifications

| **Series Specification** | **Test Statistic** | **1% Critical Value** | **5% Critical Value** | ***p*-value** |
|---|---|---|---|---|
| Raw Series $Y_t$ (Constant) | $-0.995$ | $-3.511$ | $-2.897$ | 0.755 |
| Raw Series $Y_t$ (Constant+Trend) | $-1.019$ | $-4.071$ | $-3.464$ | 0.942 |
| Differenced $\Delta_{12}\Delta Y_t$ (Constant) | $-5.886$ | $-3.521$ | $-2.901$ | $< 0.001$ |

### A.2 Residual Autocorrelation Check

Figure A.1 illustrates the Ljung-Box portmanteau test $p$-values across 24 historical lags computed on the residuals of the optimized SARIMA$(1, 1, 2) \times (0, 1, 1)_{12}$ model. The fact that all coefficients remain strictly above the horizontal 5line) refutes the mathematical thesis that the remaining error sequence acts strictly as white noise, thereby confirming the presence of uncaptured serial correlation.

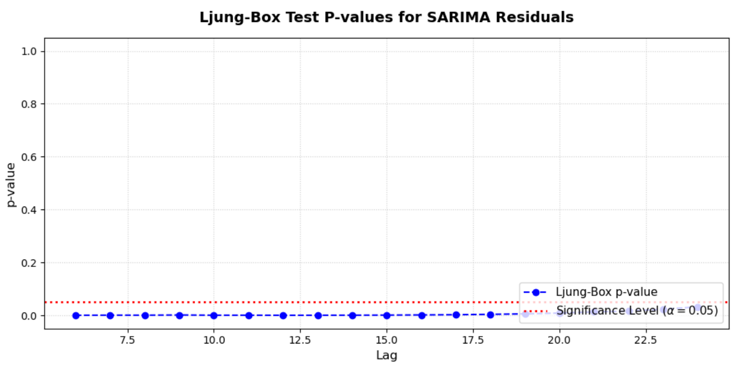


Figure A.1: Ljung-Box $p$-values across temporal lags for SARIMA residuals.

# B Chronological Context of the 2023 Wildfire Evacuation Shock

To provide qualitative support for the empirical error inflation interpreted in Section 5, Table B.1 outlines the key administrative milestones of the force majeure event that disrupted the single-access lifeline corridor of Highway 3.

Table B.1: Timeline of the 2023 Yellowknife Wildfire Emergency Impacting Highway 3

| **Date (Year 2023)** | **Administrative Action and Traffic Corridor Dynamics** |
|---|---|
| August 13 | Wildfires approach within 30 km of Yellowknife; structural alerts activated. |
| August 15 | Local state of emergency declared; localized highway closures due to heavy smoke. |
| August 16 (Evening) | Comprehensive mandatory evacuation order issued for the entire capital population ($>$ 20,000 residents). |
| August 17–19 | Mass overnight vehicular outflow at Counter Station 3-026; near-physical highway capacity saturation. |
| August 20 | Evacuation finalized; Highway 3 placed under strict lockdown restricting civilian access. |
| September 6 | Evacuation order officially downgraded to an alert status; staggered re-entry flow begins. |

# C Computational Environment and Replication Framework

To guarantee absolute scientific transparency and facilitate the exact replication of the empirical forecasting performance metrics, the full programmatic pipeline is open-sourced.

## C.1 Software Specifications

The entire predictive modeling workflow was executed on a standardized computational platform utilizing the Python programming language platform (Version 3.10.12). The core algorithmic dependencies and mathematical libraries are locked into the following specific distribution versions:

- `statsmodels` library (Version 0.14.0) — deployed for ADF testing and SARIMA grid search execution.
- `scikit-learn` framework (Version 1.3.0) — utilized for Holt-Winters smoothing parameters optimization.
- `xgboost` regressor engine (Version 1.7.6) — operated for regularized gradient boosted decision tree ensemble training.

## C.2 Open-Source Repository

The complete Jupyter Notebook configuration file (`monthly_tsa_2.ipynb`), raw multi-year administrative data streams, data preprocessing matrices, and visualization source scripts are hosted publicly. Researchers and transport planners can clone or inspect the complete architecture directly via the following dedicated GitHub repository:

https://github.com/t-man-nd/Highway-Volume-Prediction